

Medical Radiology and Imaging

Student Handbook and Module Information Booklet

Bachelor of Medicine, Bachelor of Surgery

Year 5 | Semester 9

Use this handbook to prepare for your radiology posting, organise your learning in the department, track the required competencies and complete your assessments. Bring it alongside your logbook and the current cohort timetable.

Module at a glance	Details
Module code	MED63500 (Clinical Rotation 5)
School	School of Medicine, Taylor's University
Posting	Two weeks at Hospital Sungai Buloh (HSB)
Credit value	4.0 credits
Prerequisite	Successful completion of Year 4 of the MBBS programme
Module leader	Assoc Prof Dr Eric Chung
Assessment	SBA/MCQ 40% OSPE 50% Written assignment 10%
Attendance	Minimum 80%
Written assignment	One submission at the end of posting (EOP)

Before your first session

- Check TiMeS for the reporting arrangements, cohort timetable and assessment notices.
- Review basic radiological anatomy and the competency list in Section 4.
- Bring your logbook, identification/name tag and the required clinical attire.
- Know your assigned modality group and identify the supervising staff member on arrival.

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1 Module overview

This posting develops the knowledge, image interpretation skills and professional behaviours needed to use diagnostic imaging in patient care. You will correlate clinical information with images, discuss appropriate investigations, and communicate with patients, referring clinicians and radiologists.

Radiology teaching begins in Phase I and continues through integrated clinical teaching in Phase II. The dedicated Year 5 posting consolidates imaging principles, radiological anatomy, recognition of common diseases and emergencies, and radiation and contrast-medium safety.

What you should gain from the posting

- A systematic approach to reviewing images and correlating findings with the clinical question.
- An understanding of the roles and limitations of common imaging modalities.
- Experience discussing investigation choices and imaging findings with a supervisor.
- Awareness of patient preparation, safety, consent, communication and professional conduct.
- A record of learning that identifies what you can explain, what you have practised under guidance and what needs further work.

How to use this handbook

Read the learning outcomes and competency levels before the posting. During each session, select relevant learning targets, record the encounter in your logbook and ask for feedback. Use the image discussion prompts in Section 5 to structure your preparation. Review your progress at mid-posting and address gaps before the end-of-posting assessment.

The modality rotations and daily programme are examples. Follow the current cohort timetable and instructions from the module team for the actual sequence of sessions, assessment venue and examination arrangements.

A suggested two week learning routine

Stage	Your focus
Before the posting	Read the handbook and review normal anatomy, modality names and basic safety concepts.
First week	Become familiar with departmental workflow; observe modality-specific preparation and procedures; practise structured image discussion under guidance.
Mid-posting	Review the logbook with a supervisor, identify missed competencies and plan focused practice.
Second week	Consolidate common and emergency imaging topics, practise explaining reasoning, and complete the written assignment.
End of posting	Complete the SBA/MCQ and OSPE, submit the written assignment and logbook, and use feedback to plan further learning.

2 Module Learning Outcome (MLO)

MLO	Learning outcome	PLOs
MLO1	Obtain a good history from patients/attendee OR clinical record attending for Radiology investigations to correlate with imaging findings to derive a correct diagnosis.	1, 3, 5
MLO2	Perform an adequate and appropriate physical examination wherever necessary in a proper sequence adopting moral, ethics and professionalism. Skillfully correlate them with correct interpretation of Radiology features to make a good diagnosis and differential diagnoses.	2, 3, 8
MLO3	Discuss with the referring doctor, plan and organize Radiology investigations and facilitate the attending doctors to manage patient's problem appropriately	2, 3, 7

Programme Learning Outcomes (PLO)

PLO	Programme learning outcome
1	Competence in clinical management of health problems and promotion of health based on the scientific and sociological principles of medicine and good clinical skills
2	Critical, constructive, evidence-based and patient-centred thinking in the delivery of effective healthcare
3	Proficiency at effective communications with relevant stakeholders in the practice of Medicine
4	Life-long learning skills as a reflective medical practitioner and researcher
5	Ability to recognise own attitudes and regulate them for optimum effect in health care delivery
6	Ability to work towards the well-being of patients and society through leadership and by forging mutually beneficial and respectful relationships
7	Ability to derive solutions that deliver the best possible health care with the available resources
8	Professionalism in health care delivery, following medical ethics and with awareness of the diversity of populations

Professional and transferable skills

Across the posting, develop clinical reasoning, communication, reflective learning, ethical decision-making and teamwork. Review your own performance, recognise uncertainty, seek supervision and use feedback to improve. These habits support the wider programme outcomes, including lifelong learning, self-regulation and collaborative care.

3 Teaching team and communication

Contact the module leader about learning or assessment matters and the programme coordinator about programme-level arrangements. For a concern arising during a clinical session, inform the supervising staff member promptly.

Name and role	Institution	Email and telephone
Dr Lim Su Yin (Neurology) (Programme Coordinator)	Taylor's University School of Medicine	Suyin.Lim@taylors.edu.my 013-8482471
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Dr Emy Saera bt Rosani (Radiology)	HSB	dr.emysaera@moh.gov.my; emysaera@yahoo.com.my 012-3750751
Dr Mohd Nazlin bin Kamarudin (Radiology)	HSB	zircon85_kn@yahoo.com 019-7574639
Dr Zulkifli Zaki bin Abdul Ghani (Radiology)	HSB	dr.zulkiflizaki@moh.gov.my 019-9313255
Dr Sarah Alicia (Radiology)	KKM	alicia.sarah@gmail.com 016-6267514
Dr Sumithra S. Ranganathan (Radiology)	Prince Court Medical Centre	sumithra.ranganathan@princecourt.com 012-6768840
Dr Tee Yu Jin (Radiology)	Not specified	virgoalan14@hotmail.com 016-3404634
Prof Yeong Chai Hong (Medical Physics)	Taylor's University School of Medicine	ChaiHong.Yeong@taylors.edu.my 016-7016875
Dr Francine Chai (Emergency Medicine)	Taylor's University School of Medicine	Francine.Chai@taylors.edu.my 017-8886691

Communicating effectively

- Use your university email for routine academic correspondence. Include your name, cohort and a clear subject line.
- Check the timetable and assessment notice before asking about arrangements. Raise unresolved questions early.
- Use the approved clinical logbook for patient details. Do not include patient-identifiable material in routine email, presentation slides or the reflective assignment.

4 Core radiology competencies

The following 37 tasks and their assigned levels are taken from Standards for Undergraduate Medical Education, Second Edition (2022), Section 4, Discipline: Radiology, printed pages 92–94. Use the reference codes to organise your learning and logbook review.

Competency levels

Level	Descriptor
1	Able to describe the imaging investigation
2	Identify the clinical indication(s) of the investigation
3	Able to justify the optimal imaging modality required for that particular clinical scenario under guidance.
4	Able to interpret the investigation under guidance.

The expected level varies by task. Levels 3 and 4 require guidance; they do not imply independent reporting or independent clinical decision-making. Ask your supervisor to help you identify learning opportunities appropriate to each level.

Principles of Radiological Techniques and Radiation Safety

Ref	Required task	Level
R1	Communicate the procedure of common radiological methods to the patient	2
R2	Identify basic radiological anatomy	3
R3	Request for appropriate investigations and have knowledge of the respective indications	3
R4	Understand the basics of radiation safety and basic contrast medium safety	3

Chest

Ref	Required task	Level
C1	Basic normal radiological anatomy of heart and lungs	3
C2	Basic understanding of main imaging techniques (radiography, CT and MRI) in thoracic imaging	2
C3	Pneumothorax	4
C4	Pleural effusion	4
C5	Lung/lobar collapse	3
C6	Lung consolidation	3
C7	Heart failure	3

Abdomen

Ref	Required task	Level
A1	Basic normal radiological anatomy of abdomen	3
A2	Basic understanding of main imaging techniques (radiography, ultrasound and CT) in abdominal imaging	2
A3	Pneumoperitoneum	4
A4	Acute abdomen - traumatic or non-traumatic	3
A5	Bowel obstruction - mechanical and paralytic ileus	3
A6	Common causes of abdominal calcification	3
A7	Foreign body	3

Musculoskeletal

Ref	Required task	Level
M1	Basic normal radiological anatomy of musculoskeletal system	3
M2	Basic understanding of main imaging techniques (radiography, CT and MRI) in musculoskeletal imaging	2
M3	Skeletal and spinal fractures	3
M4	Skeletal joint dislocation / spinal dislocation	3
M5	Bone and soft tissue infection	1
M6	Bone and soft tissue tumour	1

CNS

Ref	Required task	Level
N1	Basic normal radiological anatomy of CNS	2
N2	Basic understanding of main imaging techniques (radiography, CT and MRI) in CNS imaging	2
N3	Infarct	3
N4	Intracranial bleed / Trauma	4
N5	Infection and tumour	1

Paediatrics

Ref	Required task	Level
P1	Basic normal understanding of main imaging techniques (radiography, CT and MRI) in paediatric imaging	2
P2	Radiation protection in children	2
P3	Non-accidental trauma	1
P4	Basic imaging features of the most common disorders of the brain, spine, chest, abdomen, and musculoskeletal system in neonates, infants, children and adolescents	1

Breast imaging

Ref	Required task	Level
B1	Basic normal understanding of main imaging techniques (mammogram and ultrasound) in breast imaging	1

Genitourinary Imaging

Ref	Required task	Level
G1	Basic normal radiological anatomy of genitourinary system	3
G2	Basic understanding of main imaging techniques (radiography, ultrasound, CT and MRI) in genitourinary imaging	2

Head and Neck Imaging

Ref	Required task	Level
H1	Basic understanding of main imaging techniques (radiography, ultrasound, CT and MRI) in head and neck imaging	1

Additional module learning

The posting also includes learning beyond the tasks listed on these three standards pages. This supports understanding of departmental practice and integration with other clinical disciplines.

- Imaging physics: X-ray production, fluoroscopy, image intensifiers, digital and computed radiography, ultrasound and Doppler, CT, MRI, mammography and bone densitometry.
- Patient preparation and radiological procedures; angiography and interventional radiology.
- Obstetric and gynaecological imaging, oncological imaging, brain tumours, and pulmonary and extrapulmonary tuberculosis.
- Nuclear medicine: gamma camera principles, 18F-FDG PET and PET-CT, 99mTc, and the role of common examinations and tracers.

- Practical learning: film interpretation, referral dialogue, FAST ultrasound demonstration and supervised hands-on practice, and the listed lumbar puncture demonstration.

5 Learning during the posting

Departmental observation

Use each modality session to connect the patient's clinical question with the investigation being performed. Identify the supervisor, ask what you may observe or participate in, and respect patient care priorities. Record what you learned and obtain the required logbook verification.

Before the examination	During observation	After the examination
Review the clinical question, relevant history and available information.	Observe how the patient is identified, informed and prepared.	Discuss how the examination addressed the clinical question.
Identify the modality and the learning targets for the session.	Ask how safety checks, positioning and image acquisition are managed.	Review representative images with the supervisor.
Ask which parts of the session you may join.	Note the terminology used to describe anatomy and findings.	Record the encounter, feedback and one further learning action.

Questions to explore in each modality

Modality	Learning questions
Radiography and fluoroscopy	What is the clinical question? Which view or procedure is being used? How are image quality, normal anatomy, patient preparation and radiation protection considered?
Ultrasound and Doppler	What is the target organ or question? How does the operator choose views? What preparation and examination limitations are discussed?
CT	Why was CT selected? Which region and protocol are being used? What are the relevant preparation and safety checks? How are the images reviewed systematically?
MRI	Why was MRI selected? What screening and preparation are required? Which sequences or planes are demonstrated, and what are their roles?
Mammography and breast ultrasound	What is the role of each technique? How are the examination and patient preparation explained? What should an undergraduate recognise at the required introductory level?
Angiography and interventional radiology	What is the indication and intended procedure? How are consent, preparation, teamwork and aftercare organised? What may a student safely observe?

A structure for discussing an image

- State the clinical question and relevant history.
- Identify the modality, body region, projection or plane, and any limitations noted by the supervisor.
- Describe the visible anatomy and findings systematically before giving an interpretation.
- Relate the findings to the clinical information and explain your reasoning.
- State your uncertainty and discuss the interpretation, investigation choice or next learning step with the supervisor.
- Summarise the key learning point and record feedback.

Use this structure as a learning aid. The supervising clinical team remains responsible for patient assessment, reporting and management.

Lectures and small group learning

Lectures are typically 60 minutes and cover imaging principles, anatomy, techniques and safety. Prepare from the relevant competency list, participate actively and review the subject afterwards. Small group tutorials and fixed learning modules use images, clinical information and exhibits to consolidate learning. Case and image discussions remain part of teaching.

Seminars

Interactive seminars are held weekly, with each session lasting 2.5–3 hours. Each seminar has clearly defined learning outcomes and focuses on key radiology topics or clinical cases. Presentations may use audiovisual materials, including clinical images, videos and PowerPoint slides.

The radiologists select topics according to students' learning needs and communicate them to the student leader several weeks in advance. Nominated students prepare and present the assigned topics, with faculty members from relevant disciplines serving as facilitators or resource persons.

The student leader coordinates preparation and maintains a presentation roster to ensure that every student participates. All students are expected to contribute actively by asking questions, discussing findings and engaging with their peers and facilitators.

Independent learning and feedback

Use independent learning time to consolidate anatomy, review the images and concepts discussed during sessions, prepare for assessments and develop your written assignment. Directed topics may be assessed. At the end of each day, identify one concept you understand better and one question to revisit. At mid-posting, review gaps with a supervisor and agree how to address them.

6 Timetable and modality rotations

These examples show how the posting can be organised. The current cohort timetable on TIMeS provides the actual allocation. Groups may divide themselves among available modalities as directed by the department; the five-group tables below are examples rather than fixed cohort requirements.

Time	Example activity
08:00–11:00	Departmental modality rotation and scheduled conferences at HSB.
11:00–13:00	Lectures, physics teaching and image or clinical discussion. EOP examinations occur towards the end of week 2.
13:00–14:00	Lunch break.
14:00–15:00	Interactive teaching, referral discussion, quizzes, fixed learning modules or independent study.
15:00–17:00	Neuroradiology, clinical skills, seminar, case discussion, examination briefing or independent study.
Integrated block	A Friday 14:30–17:30 lecture/seminar block may appear in the published timetable.

Week 1 example rotation

Modality	Mon	Tue	Wed	Thu	Fri
Radiography, Mammography, Fluoroscopy	A	B	E	C	D
MRI scan	B	C	A	D	E
CT scan	C	D	B	E	A
Angiography / Interventional Radiology	D	E	C	A	B
USS	E	A	D	B	C

Week 2 example rotation

Modality	Mon	Tue	Wed	Thu	Fri
Radiography, Mammography, Fluoroscopy	E	A	B	D	C
MRI	A	B	C	E	D
CT	B	C	D	A	E
Angiography / Interventional Radiology	C	D	E	B	A
USS	D	E	A	C	B

Groups A–E rotate through the modalities shown. USS denotes ultrasound. Attend departmental conferences when scheduled and follow changes communicated by the module team.

7 Assessment and completion

Assessment comprises three summative components. The percentages below refer to the overall EOP mark. Formative learning and feedback take place throughout the posting.

Component	Overall weight	Timing
Single best answer (SBA)	40%	End of posting
Objective structured practical examination (OSPE)	50%	End of posting
Written reflective assignment	10%	One submission at the end of posting

Single best answer (SBA)

The EOP theory test comprises **20 single best answer questions**, each with **four answer choices** and contributes **40% of the overall end-of-posting mark**. Each correct answer gains one mark and there is no negative marking. Questions assess knowledge, analysis, clinical interpretation, critical thinking and medical ethics. Prepare across the core competencies and the scheduled teaching content.

Objective structured practical examination (OSPE)

The OSPE comprises **three questions** and contributes **50% of the overall end-of-posting mark**. It assesses image interpretation, focused diagnostic reasoning, advice and communication. Images may include CT, MRI, ultrasound and plain radiographs. The examination briefing will specify the format and time allocated to each question.

To prepare for the OSPE, students should:

- Practise describing imaging findings in a clear, systematic sequence rather than naming a diagnosis alone.
- Relate the findings to the clinical information provided and explain their diagnostic reasoning.
- Recognise uncertainty and the limitations of their interpretation.
- Use feedback from teaching sessions to identify learning gaps and guide further practice.

Written reflective assignment

Submit one written assignment based on a seminar topic or an interesting clinical case at the end of the posting. This component contributes 10% of the overall EOP mark. Assessment is of the written submission only; no oral presentation is required for this component.

Format: Microsoft Word, Times New Roman 12-point font, 1.5 line spacing and 1,000–3,000 words. The assessment notice will specify the submission route and exact deadline. Use relevant, appropriately de-identified images when they support the discussion and provide references for sources used.

Suggested assignment structure

Part	What to include
Introduction	Introduce the topic or de-identified clinical context and explain its relevance to radiology.

Part	What to include
Clinical question and investigations	Explain the relevant clinical information, imaging investigations and their clinical relevance.
Systematic image discussion	Describe relevant anatomy and findings in a clear sequence. Explain how they support or limit interpretation.
Reasoning and discussion	Discuss the diagnosis or differential diagnoses and relevant management principles within the scope of the selected topic.
Reflection and learning	Explain what you learned, where you were uncertain, how feedback or reading changed your understanding, and what you will do next.
Conclusion and references	Summarise the principal learning points and cite the sources used consistently.

Written assignment rubric

Criterion	Maximum marks	Marks awarded
Introduction and patient data	5	
Appropriateness of structure and content	5	
Clarity of presentation of the subject and data	10	
Radiological investigations, clinical relevance and systematic examination of the images	20	
Image interpretation and summary	20	
Discussion of diagnosis, provisional diagnosis and differential diagnoses	10	
Principles of management	10	
Ability to stimulate interest and further exploration	10	
Summary and learning issues	5	
Appropriateness of sources and references	5	
Total	100	

Formative assessment

Supervisors use the logbook, clinical participation, attendance and professional behaviour to review progress and provide feedback. Present your logbook at mid-posting and at the end. Use feedback to identify learning gaps and document how you addressed them.

Passing requirements

You must pass both the SBA and OSPE component. Each component requires at least 50% to pass the overall module.

Component	Calculation and pass requirement
SBA	SBA/MCQ contributes 40 overall marks and the assignment contributes 10. Pass: at least 25 out of these 50 marks, equivalent to 50% of the SBA component.
OSPE	OSPE contributes 50 overall marks. Pass: at least 25 out of these 50 marks, equivalent to 50% of the OSPE component.

An overall average does not replace the requirement to pass each component separately. You must also achieve at least 80% attendance and complete the required logbook and coursework.

Remediation and resit

Students who fail the EOP assessment may attend remedial classes determined by the discipline coordinator and Phase II programme director and resit the failed component. Resit marks are not included in the final aggregate for the end-of-year examination. Deficiencies in attendance or required coursework may be remedied during semester breaks under the posting coordinator's guidance.

8 Professional conduct and logbook

Attendance and participation

Attend scheduled placements and teaching sessions punctually. A minimum of 80% attendance is required. Attendance is monitored by the student leader and subleader and verified by the supervising clinician or nursing staff. If you cannot attend or encounter a problem with your allocation, inform the module team promptly and follow the programme's reporting process.

Conduct in the clinical setting

- Demonstrate respect, teamwork and active participation. Dress formally, wear the required white coat and display your name tag.
- Introduce yourself as a medical student and obtain verbal consent before taking a history, examining a patient or participating in a procedure. Follow the clinical setting's chaperone requirements.
- Follow departmental safety instructions and take part only within the scope authorised by the supervisor. Ask before entering controlled areas or handling clinical equipment.
- If you are uncertain or concerned about a patient, image or procedure, alert the supervising staff member. Do not present an unverified student interpretation as a formal radiology report.
- Respect confidentiality. Keep patient information in the approved clinical record or logbook and follow the institution's storage and access arrangements.

Completing the logbook

Bring the logbook throughout the posting, keep it tidy and include the required student photograph. Enter the required details on the same day and obtain a dated signature from the supervising medical officer or radiologist. Present the logbook at mid-posting and submit it at the end of the posting.

Record	Details to include
Patient and encounter	Date, patient name and registration number, and modality or session.
Clinical information	Relevant clinical data, indication for the investigation and diagnosis or provisional diagnosis.
Learning activity	Investigation or procedure observed, image discussion, seminar, conference or teaching-file work.
Learning and feedback	Relevant competency reference, principal learning point, feedback and follow-up learning action.
Verification	Date and signature of the supervising medical officer or radiologist.

Patient names and registration numbers belong in the approved clinical logbook as required for this posting. Keep them out of the reflective assignment, ordinary email and personal study notes. Do not share logbook pages or patient images through public or personal channels.

Mid posting review

- Check that entries and attendance are up to date and verified.
- Review progress across the required modalities and core competencies.
- Identify cases or topics not yet encountered and arrange suitable teaching-file or supervised discussion opportunities.

- Record feedback and a short plan for the remaining sessions.

End of posting checklist

- Required attendance achieved and recorded.
- Logbook completed, reviewed and submitted.
- One written reflective assignment submitted.
- SBA/MCQ and OSPE completed.
- Outstanding feedback and any remedial arrangements understood.

9 Student learning time

Student learning time (SLT) includes scheduled learning, independent study and assessment preparation. The module carries 4.0 stated credits. Two learning-time views are provided below: the detailed activity allocation and the curriculum-format SLT table.

Detailed allocation

Topic or activity	Delivery	Contact hours	IL hours	Total hours
Module introduction, posting briefing and HSB modality orientation	Lecture and discussion	1	2	3
Ionising radiation in X-ray, CT, nuclear medicine and mammography	Lecture and discussion	1	2	3
Physical principles of ultrasound and Doppler sonography	Lecture and discussion	1	2	3
Applied radiological anatomy on ultrasound, CT and MRI	Lecture and discussion	1	2	3
Radiological investigations and procedures, preparation and precautions	Lecture and discussion	1	2	3
MRI principles, applications and safety	Lecture and discussion	1	2	3
Radiation hazards and radiation safety	Lecture and discussion	1	2	3
Imaging of renal diseases	Lecture and discussion	1	2	3
Hepatobiliary imaging	Lecture and discussion	1	2	3
Imaging in obstetrics	Lecture and discussion	1	2	3
Imaging of spinal disorders and trauma	Lecture and discussion	1	2	3
Imaging in gynaecology	Lecture and discussion	1	2	3
Interventional radiology parts 1 and 2	Lecture and discussion	2	4	6
Neuroimaging in chronic brain diseases	Lecture and discussion	1	2	3
Coin lesion on chest radiography	Lecture and discussion	1	2	3
General principles of oncological imaging	Lecture and discussion	1	2	3
Imaging of brain tumours	Lecture and discussion	1	2	3
Radiology in ORL (Semester 6)	Integrated lecture	1	2	3
Principles of orthopaedic imaging (Semester 7)	Integrated lecture	1	2	3
Imaging in chest trauma	Seminar	3	6	9
Disease detection	Seminar	3	6	9
Child with abdominal distension (Semester 7)	Integrated seminar	2	4	6

Topic or activity	Delivery	Contact hours	IL hours	Total hours
Pulmonary and extrapulmonary tuberculosis (Semester 5)	Integrated seminar	2	4	6
Stroke (Semesters 5 and 6)	Integrated seminar	2	4	6
Interactive OSPE and clinical correlation	Clinical skills	2	1	3
Interactive image discussion for referral	Clinical skills	2	1	3
FAST ultrasound demonstration and hands-on practice	Clinical skills	2	1	3
Film interpretation	SGD/FLM	2	2	4
Quiz 1 Orthopaedic imaging in adults and children	Quiz	2	4	6
Quiz 2 Chest radiology	Quiz	2	4	6
Clinical teaching with ten sessions of three hours	Modality room	30	10	40
Write-up with one theme or title	Not specified	-	4	4
Self-directed and directed learning	Independent study	-	6	6
End-of-posting assessment	Not specified	3	3	6
Total		77	100	177

Integrated topics retain their semester labels. IL denotes independent learning. A dash indicates that no contact hours are allocated in that row.

Curriculum format SLT table

The grouped headings and column order below match the supplied SLT snapshot. All values are hours. Blank online cells indicate that no synchronous online hours are allocated in this view. Row totals are calculated from the activity columns.

Course Content Outline and Subtopics		CLO*	Learning and Teaching Activities**								Total SLT	
			Face-to-Face (F2F)									NF2F Independent Learning (Asynchronous)
			Physical				Online/Technology-mediated (Synchronous)					
			L	T	P	O	L	T	P	O		
1	Briefing on radiology modalities available in the Department of Radiology	3	1		1						1	3
2	Applied radiological anatomy	1, 2	1								1	2
3	CT scan (O = posting in the Radiology Department)	1, 2	1		1	2					3	7
4	MRI (O = posting in the Radiology Department)	1, 2	1		1	2					3	7
5	Contrast studies / fluoroscopy / hysterosalpingography (O = posting in the Radiology Department)	1, 2	1		1	2					3	7
6	Interventional radiology (O = posting in the Radiology Department)	1, 2	1		1	2					3	7
7	Ultrasonography / X-ray / mammography (O = posting in the Radiology Department)	1, 2	1		1	2					3	7
8	Fundamental physics of radiology imaging	1, 2	5								6	11
9	Radiology in the paediatric age group (O = interdepartmental conference)	1, 2	2			1					2	5
10	Images for referral	1, 2	2								3	5
11	Radiology of common medical and surgical diseases	2, 3	2	2		3					3	10
12	Radiology in obstetrics and gynaecology (O = interdepartmental conference)	2, 3	2			1					2	5

Course Content Outline and Subtopics		CLO*	Learning and Teaching Activities**								Total SLT	
			Face-to-Face (F2F)									
			Physical				Online/Technology-mediated (Synchronous)					NF2F Independent Learning (Asynchronous)
L	T	P	O	L	T	P	O					
13	Radiology of musculoskeletal trauma and diseases (O = interdepartmental conference)	2, 3	2			1					2	5
14	Neuroradiology (O = interdepartmental conference)	2, 3	2			1					2	5
15	Oncological radiology (O = interdepartmental conference)	2, 3	2			1					2	5
Teaching and learning subtotal			26	2	6	18	0	0	0	0	39	91

* CLO = course learning outcome. ** L = lecture; T = tutorial; P = practical; O = other learning activity, including the departmental postings and conferences identified in the topic column. NF2F = non-face-to-face learning.

Assessment allocation for the curriculum format view

Assessment	Weight	Physical hours	Online hours	NF2F hours	Total SLT
Written reflective assignment	10%	0	0	4	4
SBA/MCQ	40%	0.5	0	1.5	2
OSPE	50%	0.5	0	1.5	2
Assessment subtotal		1	0	7	8
Teaching and learning subtotal		52	0	39	91
Grand total for this view		53	0	46	99

10 Learning resources (Optional)

Use introductory texts to establish core concepts, then consolidate them with image cases, teaching files and feedback from departmental sessions. Focus on the competencies and scheduled topics in this handbook. Keep organised notes and revisit areas where you needed guidance.

Books

- Balasubramaniam R, Watson N. Chapman & Nakielny's Guide to Radiological Procedures. 8th ed. Elsevier; 2023. Print ISBN 9780323933636; eBook ISBN 9780323933872.
- Novelline RA. Squire's Fundamentals of Radiology. 6th ed. Harvard University Press; 2004. ISBN 9780674012790.
- Goodman LR. Felson's Principles of Chest Roentgenology: A Programmed Text. 5th ed. 2021. ISBN 9780323625678.
- Anatomy for Diagnostic Imaging. 4th ed. Elsevier; 2025. ISBN 9780443105609.
- Core Radiology: A Visual Approach to Diagnostic Imaging. Volumes 1 and 2. Cambridge University Press; 2021. ISBN 9781108984454.
- Qureshi Z, editor. The Unofficial Guide to Radiology. 2nd ed. Elsevier; 2024. ISBN 9780443109140.
- Qureshi Z. The Unofficial Guide to Radiology: 100 Practice Abdominal X Rays with Full Colour Annotations and Full X Ray Reports. 1st ed., 2020, Cambridge University Press. ISBN 9781910399033.
- Pocketbook Guide to Radiographic Image Evaluation. 2nd ed. Penerbit Press UiTM; 2019. ISBN 9789673632237.
- Herring W. Learning Radiology: Recognizing the Basics. 4th ed. Elsevier; 2019. ISBN 9780323567299.
- Sutton D. Textbook of Radiology and Imaging. Elsevier, December 2023. ISBN 9788131259597.
- Howlett DC, Ayers B. The Hands-On Guide to Imaging. Wiley; 2004. ISBN 9781405115513.
- Brant WE, Helms C. Fundamentals of Diagnostic Radiology. 4th ed. Lippincott Williams & Wilkins; 2015.

Online learning and journals

- Learning Radiology — <http://www.learningradiology.com/>
- Radiography Online — <http://www.radiographyonline.com/>
- Radiology Education — <http://www.radiologyeducation.com>
- Radiopaedia — <https://radiopaedia.org>
- Nuclear medicine teaching file — <https://www.mir.wustl.edu/patient-care/clinical-specialties/nuclear-medicine/teaching-file>
- Radiology — <https://pubs.rsna.org/journal/radiology>
- European Journal of Radiology — <https://www.ejradiology.com>
- American Journal of Roentgenology — <https://www.ncbi.nlm.nih.gov/labs/journals/ajr-am-j-roentgenol/>

Standards reference

Standards for Undergraduate Medical Education. Second Edition. 2022. Section 4, Core Competencies, Discipline: Radiology, printed pages 92–94 (PDF pages 111–113 in the supplied file).

Abbreviations

Abbreviation	Meaning
MLO / PLO	Module / programme learning outcome
CNS / CT	Central nervous system / computed tomography

Abbreviation	Meaning
CSS / FLM	Clinical skills session / fixed learning module
DSL / SDL / SGD	Directed learning / self-directed learning / small group discussion
EOP / SLT / IL	End of posting / student learning time / independent learning
HSB / TCS	Hospital Sungai Buloh / Taylor's Clinical School
MRI / NM	Magnetic resonance imaging / nuclear medicine
ORL / PACS	Otorhinolaryngology / picture archiving and communication system
OSPE	Objective structured practical examination
SBA / MCQ	Single best answer / multiple choice question
US / USS / FAST	Ultrasound / ultrasound scan / focused assessment with sonography for trauma